Analysis of Ranging Precision in an FMCW Radar Measurement Using a Phase-Locked Loop

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***Abstract*— The standard deviation in a frequency modulated continuous wave radar distance measurement using a charge pump phase-locked loop (PLL) is calculated analytically. The phase noise of the PLL is modeled as an Ornstein–Uhlenbeck process resulting in a Lorentzian spectrum. We calculate the distance error as a function of the receiver noise bandwidth and the target distance. Depending on the frequency estimation algorithm and the target distance, the rms distance error due to PLL phase noise increases by about 6–9 dB with doubling the target distance. By contrast, the white noise in the radar receiver raises the distance error by about 12 dB in the far field with distance doubling, making this error contribution dominant for large target distances. These findings are verified by measurements on a scalable 61/122-GHz radar sensor platform.**

***Index Terms*— Fractional-N synthesizer, frequency modulated continuous wave (FMCW), radar, ranging precision, phase noise, jitter. EDICS: ACS170, ACS180, ACS280, ACS300.**

I. Introduction

FREQUENCY synthesis in frequency-modulated continuous-wave (FMCW) radar can be performed by a direct digital frequency synthesizer driving an integer-N phase-locked loop (PLL) [1]. As an alternative, a fractional-N PLL can be used in conjunction with a constant input frequency as the FMCW generator [2]–[10]. For both cases, PLL phase noise may limit the ranging precision of the radar system. A lower bound for the ranging precision of primary radar systems was given in [10] and [11]. However, the measured ranging error was significantly higher than this lower bound. A secondary radar system was considered in [12], where an estimate for the root-mean-square (rms) frequency error was derived from the PLL phase noise spectrum.

For achieving a high ranging precision, a primary radar system has the advantage that the same voltage-controlled oscillator (VCO) can be used in the transmitter and in the receiver. The mixer in the receiver multiplies the incoming signal with the transmitted signal. After low-pass filtering, the mixer output phase is basically the difference between the phase of the transmitted signal at time t and its earlier value at $t - t_d$. As a result, much of the phase noise is canceled in such a homodyne radar system. In this analysis, we consider a homodyne radar system as it was realized in [9].

In addition to phase noise, the thermal noise in the receiver antenna contributes to the distance error. The corresponding signal-to-noise ratio (SNR) at the antenna is further reduced by white noise sources (shot noise, thermal noise) in the receiver, particularly in the low-noise amplifier (LNA). The effect of 1/f-noise is usually minimized by choosing a ramp slope large enough to shift the radar signal to above the 1/f-noise corner frequency. Since the signal at the radar receiver is strongly reduced with increasing target distance, we expect a strong dependence of the corresponding distance error on the target distance. An interesting question is which mechanism dominates the distance error in an FMCW radar system. This question could be answered by system simulations or by a stochastic analysis of the radar system.

Simulation results of a radar system including VCO phase noise were presented in [13] and [14]. The importance of a low VCO phase noise in a two-target scenario was highlighted. In [15] the rms phase error in an FMCW radar system using a charge pump (CP) PLL was calculated as a function of basic PLL design parameters, but range correlation effects were not included. A linear frequency-domain noise model of a homodyne radar system was presented in [11]. Here, range correlation effects were considered by a weighting function for the phase noise spectrum based on the model in [16]. This linear, stationary description of the phase noise allows an easy and efficient estimation of rms phase error and signal-to-noise ratio for many cases. However, non-stationary effects as observed in FMCW radar using fast frequency ramps cannot be described. A non-stationary phase noise model based on an Ornstein-Uhlenbeck (OU) process was employed in [17] to calculate the rms phase error in a homodyne radar system, but the rms *distance* error, i.e., the ranging precision was not considered in the analysis. A stationary frequency-domain model of the ranging precision due to PLL phase noise including range correlation effects was presented in [18]. However, receiver noise was disregarded in the calculation of the distance error preventing a sound comparison with range error measurements.

This paper presents a unified time-domain and frequency-domain description of the ranging precision in homodyne FMCW radar systems. The model includes both PLL phase noise and receiver noise. Experiments are presented to support the theoretical results.

II. Time-Domain Phase Noise Modeling

In an FMCW radar system driven by a PLL the total sweep time T_s can usually be subdivided into two phases.

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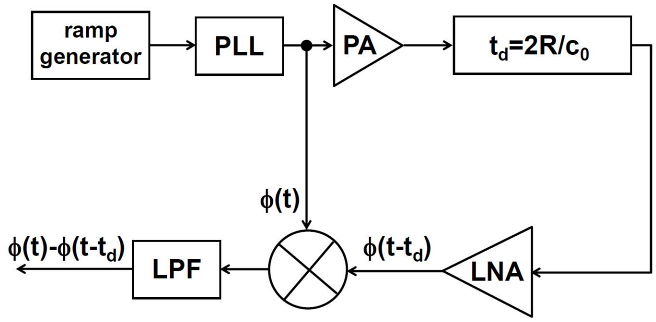

Fig. 1. Simplified block diagram of homodyne radar frontend.

For $0 < t < t_s$ the PLL settles to the desired frequency. For $t_s < t < T_s$ the PLL is in the steady state. As shown in [15], the dynamics of the phase error for $t_s < t < T_s$ is the same as in the case of a constant PLL output frequency. The only difference is the appearance of a static phase error ϕ_0 at the phase detector input in the case of a frequency ramp. This error is proportional to the ramp slope and can be minimized by using a large CP current. The settling phase $0 < t < t_s$ is usually disregarded in the evaluation of the radar signal, and the second phase $t_s < t < T_s$ is used for the estimation of the target distance. For slow ramps, the PLL settling phase is not relevant, since it represents only a small portion of the total sweep time. However, for fast ramps the settling time becomes relevant and should be minimized. A time-domain model allows to describe both the PLL settling and the phase noise in a unified framework. Next, we will briefly summarize the nonstationary time-domain phase noise model presented in [17]. Subsequently, we reformulate the OU process as a stochastic integral equation suited for numerical investigations.

Fig. 1 shows a simplified model of the radar frontend. The radar signal roundtrip time t_d is symbolized by a delay line. The mixer multiplies the transmitted signal at time t with the signal at an earlier time $t - t_d$. After the low-pass filter (LPF), the phase of the radar signal represents basically the difference of the PLL output phases at these points of time. The steady-state auto-correlation function (ACF) at the LPF output can be written as

$$R_{\text{out}}(\tau) = < \phi_{\text{out}}(t)\, \phi_{\text{out}}(t+\tau) > \tag{1}$$

where the brackets denote the stochastic average. We assume for the moment that the only effect of the radar receiver on the phase is the elimination of the high-frequency component around $2\,f_0$, where f_0 is the PLL output frequency. In this case, the receiver output phase $\phi_{\text{out}}(t)$ represents the difference between the PLL output phase of the transmitter $\phi(t)$ and a time-delayed version according to

$$\phi_{\text{out}}(t) = \phi(t) - \phi(t - t_d)\,. \tag{2}$$

For a low filter bandwidth some modifications are required, which will be discussed in Section III. By substituting (2) into (1) we obtain

$$R_{\text{out}}(\tau) = 2R(\tau) - R(\tau + t_d) - R(\tau - t_d) \tag{3}$$

where we exploited the symmetry $R(\tau) = R(-\tau)$. In order to calculate the auto-correlation $R(\tau)$ of the PLL output phase we use a stochastic model.

The phase of a VCO placed in an overdamped PLL can be described by an OU process. This process is defined by the Langevin equation [19]

$$\frac{\mathrm{d}}{\mathrm{d}t}\phi(t) + \omega_L \phi(t) = F(t) \tag{4}$$

where the ACF of the Gaussian noise force is given by

$$< F(t)F(t+\tau) > = 2D_\phi \delta(\tau)\,. \tag{5}$$

Here, $\omega_L = 2\pi f_L$ where f_L is the loop bandwidth in hertz, D_ϕ is the phase diffusivity, and $\delta(\tau)$ is the Dirac function. In the limit $\omega_L \to 0$ the OU process becomes a Wiener process describing the phase of a free running VCO having only white noise sources [20]–[22]. For the initial condition $\phi(0) \neq 0$ the OU process is a non-stationary process. This is the case, e.g., in an FMCW radar system whenever the slope of the frequency ramp changes, since the static phase error ϕ_0 is proportional to the ramp slope. As a result, the dynamic phase error $\phi(t)$ will experience a phase jump at the beginning of a frequency sweep. This value can be minimized by using a large CP current and a small loop filter capacitance [17]. For ultra-fast ramps, the required filter capacitance might be too small to guarantee PLL stability. Here, adjusting the value of the phase jump to slightly above 2π can significantly reduce the PLL settling time while keeping PLL stability.

In order to generate an OU process on a discrete time grid with a time step size $\mathrm{d}t$, we rewrite (4) as

$$\phi(t + \mathrm{d}t) = \phi(t) - \omega_L\, \phi(t)\, dt + \sqrt{2D_\phi\, \mathrm{d}t}\, N(t) \tag{6}$$

where $N(t)$ is a temporally uncorrelated zero-mean normal variable with a standard deviation of one. In order to generate the normal (Gaussian) distribution for N on a computer, the Box-Muller method can be used [23]. Equation (6) is exact for $\mathrm{d}t \to 0$ only. A numerical simulation algorithm that is exact for any time step $\Delta t > 0$ was presented in [24]. It is based on the knowledge of the analytical solution for this process. In a strict mathematical framework, (6) rather than (4) should be considered as the definition of an OU process. This is because the OU process is continuous everywhere but differentiable nowhere. In other words, (4) contains a derivative which does not even exist and should be regarded as a symbolic representation of (6).

III. Phase Noise Modeling in the Steady State

Now we consider the phase error at the receiver output. As shown in Fig. 1, the delay time $t_d = 2R/c_0$ depends on the target distance R, where c_0 is the speed of light. As derived in [17], the ACF of the steady-state phase at the receiver output reads

$$R_{\text{out}}(\tau) = \frac{D_\phi}{\omega_L} \times \Big[2\exp(-\omega_L|\tau|) - \exp(-\omega_L|\tau + t_d|) - \exp(-\omega_L|\tau - t_d|)\Big]\,. \tag{7}$$

For $t_d \to 0$ we find $R_{\text{out}}(\tau) \to 0$, corresponding to a complete phase noise cancellation. For the general case, we obtain by

Fourier transformation the phase noise spectrum given by

$$S_{\phi,\mathrm{out}}(f) = \frac{2}{\pi^2}\frac{D_\phi}{f^2 + f_L^2}\sin^2(\pi f\, t_d)\,. \tag{8}$$

This result can be written as

$$S_{\phi,\mathrm{out}}(f) = 4S_\phi(f)\sin^2(\pi f\, t_d) \tag{9}$$

where

$$S_\phi(f) = \frac{1}{2\pi^2}\frac{D_\phi}{f^2 + f_L^2} \tag{10}$$

is the PLL phase noise spectrum representing a Lorentz function which is a second-order low-pass filter. In this form, our result is a special case of equation (16) in [16] applied to the case where a PLL is used for frequency generation. Note that the PLL spectrum is band-pass filtered by the weighting function $4\sin^2(\pi f\, t_d)$ and low-pass filtered to reject the component at $2f_0$ at the mixer output. If additional low-pass filtering is relevant, then (8) must be modified accordingly [11]. This results in an expression for the phase variance at the receiver output given by

$$\sigma_\phi^2 = 8\int_0^{b_n} S_\phi(f)\sin^2(2\pi f R/c_0)\mathrm{d}f. \tag{11}$$

Here, b_n is the receiver noise bandwidth and R is the target distance. The standard deviation σ_ϕ represents the rms phase jitter in radians. Sometimes, the tracking jitter $\sigma_{tr} = \sigma_\phi/(2\pi f_0)$ is used as a PLL phase noise figure of merit [25].

Following [26], the signal-to-noise ratio SNR can approximately be calculated from

$$1/SNR = 1/SNR_0 + 2\sigma_\phi^2 \tag{12}$$

where SNR_0 is the signal-to-noise ratio in the absence of PLL phase noise. The factor of two in (12) results from the fact that, unlike in [26], we normalized the noise voltage to the effective signal amplitude rather than to the peak amplitude, see also (7) in [27].

So far, we have expressed the phase noise spectrum at the receiver output by two relatively abstract quantities, namely, the phase diffusivity D_ϕ and the loop bandwidth f_L. Now we will express these quantities and the corresponding phase noise spectrum by basic design parameters of a charge pump PLL. The phase noise of the free running VCO is usually known before the PLL design. Let us assume the VCO phase noise at a specific offset Δf is $S_{\phi,\mathrm{VCO}}(\Delta f)$. Here, Δf is an offset in the region of the phase noise spectrum with a -20 dB per decade slope, e.g., $\Delta f = 1$ MHz. According to (26) in [21] the phase diffusivity reads

$$D_\phi = 2\pi^2 S_{\phi,\mathrm{VCO}}(\Delta f)(\Delta f)^2\,. \tag{13}$$

The loop bandwidth in a strongly overdamped first-order charge pump PLL is given by [15]

$$f_L = \frac{I_{\mathrm{CP}}\, K_{\mathrm{VCO}}\, R_1}{2\pi N} \tag{14}$$

where I_{CP} is the CP current, K_{VCO} is the VCO gain in hertz per volt, R_1 is the main loop filter resistance, and N is the division ratio of the feedback divider. For a weakly damped PLL the bandwidth is half that value, but in reality the damping is typically closer to the strongly overdamped case to avoid peaking in the phase noise spectrum. Finally, the delay time can be replaced with the target distance according to

$$t_d = \frac{2R}{c_0}. \tag{15}$$

Substituting (13), (14) and (15) into (8) we obtain

$$S_{\phi,\mathrm{out}}(f) = \frac{4S_{\phi,\mathrm{VCO}}(\Delta f)(\Delta f)^2\sin^2(2\pi f R/c_0)}{f^2 + [(I_{\mathrm{CP}}\, K_{\mathrm{VCO}}\, R_1)/(2\pi N)]^2}\,. \tag{16}$$

For small offsets f or for small target distances R we obtain from the Taylor expansion of the sine

$$S_{\phi,\mathrm{out}}(f) \approx \frac{16\pi^2 S_{\phi,\mathrm{VCO}}(\Delta f)(\Delta f)^2\, f^2\, R^2/c_0^2}{f^2 + [(I_{\mathrm{CP}}\, K_{\mathrm{VCO}}\, R_1)/(2\pi N)]^2}\,. \tag{17}$$

Equations (16) and (17) are suited to calculate the phase noise spectrum of the intermediate frequency (IF) radar signal from VCO phase noise, basic PLL parameters and the target distance.

IV. Analysis of the Phase Error

In this section, we consider two special cases, where the IF radar signal and the receiver noise bandwidth b_n are clearly above or clearly below the PLL loop bandwidth.

A. Fast Ramps

Let us consider the case of fast ramps where the radar signal frequency and b_n are clearly above the PLL loop bandwidth f_L. In this case, the PLL phase noise spectrum can be approximated in the relevant frequency region by the VCO phase noise $S_{\phi,\mathrm{VCO}}(f) = S_{\phi,\mathrm{VCO}}(\Delta f)(\Delta f/f)^2$. Here, we assumed that the quantization noise in the PLL can be suppressed to a negligible level. In view of the high input frequencies in fractional-N PLLs, this can typically be achieved by using a high-order low-pass filter in the PLL. Performing the integration in (11) yields for $b_n \to \infty$

$$\sigma_\phi^2 = 8\pi^2 S_{\phi,\mathrm{VCO}}(\Delta f)(\Delta f)^2(R/c_0)\,. \tag{18}$$

For example, with a VCO phase noise of -80 dBc/Hz at 1 MHz offset and a target distance of $R = 25$ m we obtain an rms phase error of $\sigma_\phi = 0.256$, which corresponds to 15°. For a distance of $R = 1$ m the phase error is as small as 3°.

B. Slow Ramps

Next, we consider slow ramps where b_n is clearly below the loop bandwidth. Assuming that the PLL phase noise can be described by a constant in-band noise floor $S_{\phi,\mathrm{fl}}$ in the relevant frequency region, we obtain from (11)

$$\sigma_\phi^2 = 8\, S_{\phi,fl}\int_0^{b_n}\sin^2(2\pi f R/c_0)\,\mathrm{d}f. \tag{19}$$

The general expression (11) for the rms phase error and the two special cases (18) and (19) will serve as the basis for the distance error calculation.

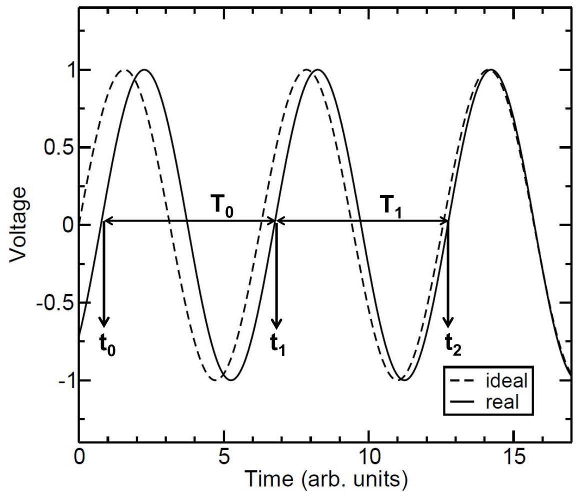

Fig. 2. Waveforms of an ideal radar output signal (dashed line) and a signal with phase noise (solid line).

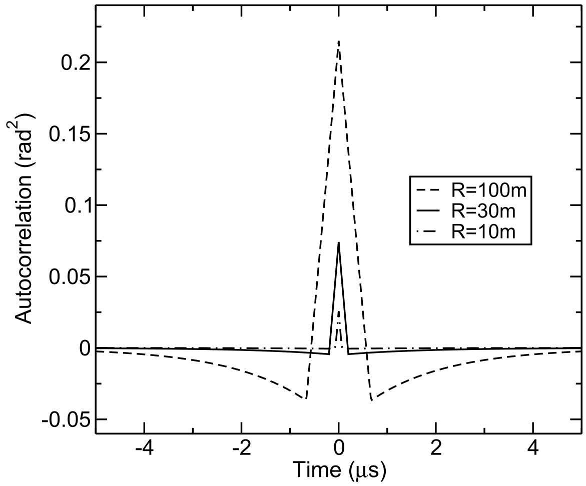

Fig. 3. Calculated ACF $R_{\mathrm{out}}(\tau)$ for a VCO phase noise of −80 dBc/Hz at 1 MHz offset and a loop bandwidth of $f_L = 100$ kHz.

V. Analysis of the Distance Error

A. General Case

In an FMCW radar system the precision of the distance measurement is related to the rms frequency error at the receiver output. From the phase error in Section IV we will calculate the frequency error and, subsequently, the rms distance error. In this paper, we will estimate the phase error from the phase noise spectrum representing the power spectral density (PSD), $S_{\phi,\mathrm{out}}(f)$, of the receiver output phase ϕ_{out}. A higher precision could be obtained by evaluating the phase of the radar signal as well [27], [28], but this is beyond the scope of our analysis.

Before calculating the rms frequency error in the presence of white noise one should carefully define the frequency. It is important to note that both the Wiener process and the OU process have the interesting property of being continuous everywhere but differentiable nowhere. This makes the definition of the frequency as the derivative of the phase problematic. Moreover, the known results from FM modulation should be used with care. Fig. 2 shows an ideal voltage signal together with a signal containing phase noise, where the phase noise is much exaggerated. In a locked CP-PLL, the mean phase of the noisy signal is the same as for the ideal signal, but the zero crossings t_i fluctuate around the ideal values. The momentary receiver output period T_i is defined as the time difference between two consecutive zero crossings of the receiver output signal as indicated in Fig. 2. Disregarding the initial time interval $0 < t < t_s \approx 10/\omega_L$ of a linear frequency sweep where the PLL settles, T_i can be considered a stationary stochastic process [15]. This implies that the stochastic average of T_i and its variance are independent of i for $t > t_s$.

We define the momentary receiver output frequency as a time-discrete quantity by

$$f_{\mathrm{out},i} = \frac{1}{T_i} = \frac{1}{t_{i+1} - t_i} \tag{20}$$

where i runs from 0 to n-1. Here, n is the number of output periods included in the evaluation of one FMCW sweep where $t_s < t < T_s$ is fulfilled. From the rms phase error we obtain the standard deviation of the corresponding timing error given by

$$\sigma_{\mathrm{t}} = \frac{T_{\mathrm{out}}}{2\pi}\sigma_\phi \tag{21}$$

where T_{out} is the ideal receiver output period. Neglecting $1/f$ noise, we find for the standard deviation of the receiver output period T_i

$$\sigma_{\mathrm{T}} = \sqrt{2}\,\sigma_{\mathrm{t}} = \frac{T_{\mathrm{out}}}{\sqrt{2}\,\pi}\sigma_\phi. \tag{22}$$

Here, we assumed that in the case of white noise sources two successive zero crossings of the output signal are statistically independent. This assumption is motivated by the observation that the phase noise spectrum of the PLL is high-pass filtered according to (11). This corresponds to a small auto-correlation time of the phase at the receiver output, since the ACF of the phase represents the inverse Fourier transform of the phase noise spectrum. Fig. 3 shows the ACF $R_{\mathrm{out}}(\tau)$ according to (7) and (13) for three different target distances.

In this example, the ACF is very small at $|\tau| \geq 1\,\mu$s for target distances up to 30 m. This justifies the assumption of uncorrelated zero crossings used in (22) as long as f_{out} is in the kHz range and the target distance is not larger than 30 m.

Assuming $\sigma_{\mathrm{T}} \ll T_{\mathrm{out}}$ the error propagation law can be applied. This yields with (22)

$$\sigma_{f_{\mathrm{out}}} = f_{\mathrm{out}}\frac{\sigma_{\mathrm{T}}}{T_{\mathrm{out}}} = \frac{f_{\mathrm{out}}}{\sqrt{2}\,\pi}\sigma_\phi\,. \tag{23}$$

One frequency sweep contains $n = T_s f_{\mathrm{out}}$ oscillations of the receiver output signal, where T_s is the sweep duration counted from reaching the steady state after PLL settling. Since $f_{\mathrm{out},i}$ is a stationary process, the best estimate for the IF frequency is the average $\overline{\mathrm{f}_{\mathrm{out}}}$ of all n frequencies within one sweep. We assume, as a first approximation, the n periods to be uncorrelated. In this case, the averaging procedure reduces the standard deviation by a factor of $1/\sqrt{n}$. Therefore, the standard deviation of this average is lower than (23) by a factor of $1/\sqrt{T_s f_{\text{out}}}$ yielding

$$\sigma_{\overline{f_{\text{out}}}} = \sqrt{\frac{2\,f_{\text{out}}}{T_s}}\,\frac{\sigma_\phi}{2\,\pi}\,. \tag{24}$$

In reality, two adjacent IF periods are not uncorrelated. Depending on the frequency estimator, this may significantly reduce the rms distance error as discussed below and in the appendix.

The receiver output frequency is related to the ramp slope $\mathrm{d}f_0/\mathrm{d}t = b_s/T_s$ and the target distance R by

$$f_{\text{out}} = \frac{b_s}{T_s}\frac{2R}{c_0} \tag{25}$$

where b_s is the sweep bandwidth in hertz. From (25) and (24) we obtain the rms distance error

$$\sigma_R = \frac{T_s\,c_0}{2\,b_s}\,\sigma_{\overline{f_{\text{out}}}} = \frac{\sqrt{T_s\,f_{\text{out}}}\,c_0}{\sqrt{8}\,\pi\,b_s}\sigma_\phi\,. \tag{26}$$

Substituting (25) into (26) we obtain the final result

$$\sigma_R = \sqrt{\frac{c_0\,R}{b_s}}\,\frac{\sigma_\phi}{2\,\pi} \tag{27}$$

where σ_ϕ is given by (11).

It is instructive to compare this result with the Cramér-Rao lower bound (CRLB) given in [11] and [10]. For $SNR \gg 1$ we find in our notation

$$\sigma_R^{\text{CRLB}} = \frac{c_0}{4\pi\,b_s\sqrt{SNR}}\,. \tag{28}$$

By combining (27), (28), (12) and (25) we obtain

$$\sigma_R = \sigma_R^{\text{CRLB}}\sqrt{n} \tag{29}$$

where $n = T_s\,f_{\text{out}} = (2b_s R)/c_0$ is the number of receiver output periods for one frequency sweep. According to the frequency definition (20) the sweep must contain at least one output period. From $n \geq 1$ and (29) we find

$$\sigma_R \geq \sigma_R^{\text{CRLB}} \tag{30}$$

as it should be. We conclude from (29) that for $n \gg 1$ the CRLB gives far more optimistic results for the standard deviation of the distance than (27). For example, for an IF frequency of $f_{\text{out}} = 100\,\text{kHz}$ and a sweep time of $T_s = 1\,\text{ms}$ we obtain $\sigma_R^{\text{CRLB}} = \sigma_R/10$. Fig. 4 shows an rms distance error measurement taken from [11] supplemented with the results of our analysis. Here, TN denotes the thermal antenna noise amplified by the receiver noise figure, PN is the phase noise contribution, QN is the system quantization noise, and MC means Monte-Carlo simulation, all presented in [11]. To obtain our analytical result (filled symbols), the phase noise spectrum measured in [11] was fitted by a fourth-order low-pass filter. Subsequently, the frequency integral over the fitted spectrum from 1 kHz to $b_n = 10\,\text{MHz}$ according to (11) was calculated numerically as a function of the target distance. From (27) we then find approximately a distance dependence $\sigma_R \propto R^{3/2}$ for small distances $R < 5\,\text{m}$, and $\sigma_R \propto R$ for $R > 20\,\text{m}$. For $R > 10\,\text{m}$ the phase-noise-related error dominates the distance error and the measured values follow the analytical results. The agreement of the measured data with the results of our analysis is much better than with the CRLB. In order to reach the CRLB, an evaluation of the phase of the IF radar signal as in [28] is indispensable. This corresponds to the inclusion of all cross-correlation terms between $f_{\text{out,i}}$ and $f_{\text{out,j}}$ for $i \neq j$ as shown in the appendix. If the auto-correlation time is much shorter than the IF period then the ACF is mainly determined by the inner part of the periods. This is also true for the phase noise spectrum representing the Fourier transform of the ACF. Therefore, a distance measurement solely based on the phase noise spectrum may underestimate the cross-correlation between different IF periods. As discussed in the appendix, the distance error, σ_R, should then be larger. This increase is especially pronounced for large target distances.

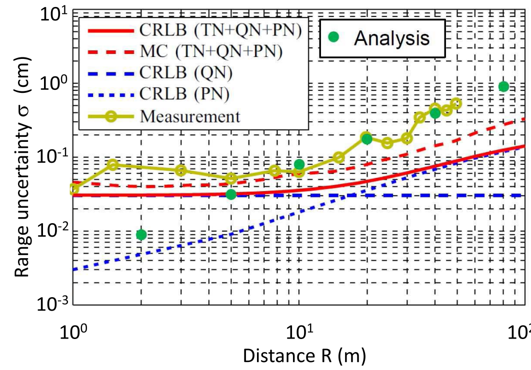

Fig. 4. Measurement results and CRLB taken from [11]. The filled circles represent our analytical result given by (27) and (11).

B. Fast Ramps

For $b_n \gg f_L$ the general result (27) can be simplified. By combining (18) and (27) we find for the variance of the distance error measurement for one sweep

$$\sigma_R^2 = \frac{2\,R^2}{b_s}S_{\phi,\text{VCO}}(\Delta f)(\Delta f)^2\,. \tag{31}$$

Remember that (31) was derived for the case $f_{\text{out}} \gg f_L$ which is more representative for fast ramps and large distances. For small distances, the more accurate expression (27) with (11) should be used.

C. Slow Ramps

By substituting (19) into (27) we obtain for the variance of the distance measurement for $b_n \ll f_L$

$$\sigma_R^2 = \frac{2\,R\,c_0}{\pi^2\,b_s}\,S_{\phi,fl}\int_0^{b_n}\sin^2(2\pi f\,R/c_0)\,\mathrm{d}f\,. \tag{32}$$

For short and medium distances we have

$$R \ll \frac{c_0}{2\pi\,b_n} \tag{33}$$

and we can replace $\sin(x)$ with x in (32). This yields

$$\sigma_R^2 \approx \frac{8}{3}\frac{R^3}{c_0\,b_s}\,S_{\phi,fl}\,b_n^3\,. \tag{34}$$

As discussed in the context of (30), $n \geq 1$ must be fulfilled to define a frequency. This results in $R \geq c_0/(2b_s)$ which limits the rms distance error to the CRLB.

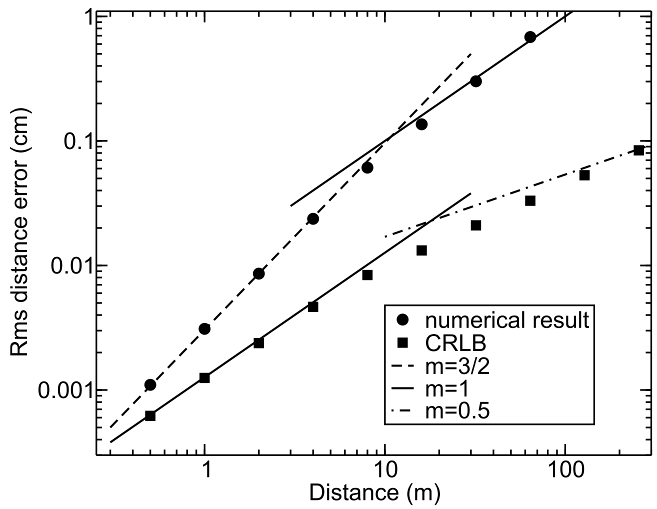

Fig. 5. RMS distance error from numerical evaluation of the phase noise spectrum (circles) and CRLB given in [10] and [11] (squares). The lines represent fitted curves according $\sigma_R \propto R^m$.

Fig. 5 shows the numerical result for the phase noise induced distance error obtained from (27) and (11). We assumed an in-band noise floor of -83 dBc/Hz, a phase noise of -115 dBc/Hz at 1 MHz offset, a loop bandwidth of 100 kHz, a noise bandwidth, b_n, of 10 MHz, and a sweep bandwidth, b_s, of 1 GHz. As evident from Fig. 5 the error is roughly proportional to R over most of the range according to (31). At small distances, however, the slope is somewhat larger as expected from (34). This effect, however, may be hidden by quantization noise in a real radar system.

VI. Analysis of Noise in the Receiver

The signal at the receiver output can be written as

$$V_{\text{out}}(t) = V_0 \cos(2\pi f_{\text{out}} t + \phi_{\text{out}}(t)) . \tag{35}$$

The signal power is $V_0^2/2$. From the signal-to-noise ratio SNR_{out} we obtain the noise power

$$\sigma_V^2 = \frac{V_0^2}{2\,\text{SNR}_{\text{out}}} . \tag{36}$$

Expressing SNR_{out} by the SNR at the receiver input and the receiver noise figure F we obtain

$$\sigma_V = V_0 \sqrt{\frac{F}{2\,\text{SNR}_{\text{in}}}} . \tag{37}$$

In order to obtain the rms timing jitter σ_t, we calculate the signal slope at the zero crossings and find from (35)

$$\left| \frac{\mathrm{d}V_{\text{out}}}{\mathrm{d}t} \right|_{V(t)=0} = 2\pi f_{\text{out}} V_0 . \tag{38}$$

Assuming $\text{SNR}_{\text{out}} \gg 1$, we can use the error propagation law and obtain from (37) and (38)

$$\sigma_t = \frac{\sigma_V}{\left| \frac{\mathrm{d}V_{\text{out}}}{\mathrm{d}t} \right|_{V_{\text{out}}(t)=0}} = \frac{1}{2\pi f_{\text{out}}} \sqrt{\frac{F}{2\,\text{SNR}_{\text{in}}}} . \tag{39}$$

Note that this procedure requires the assumption that the slope of $V(t)$ is the same as for the noiseless case. For fast and/or strong additive noise this assumption may not be fulfilled. In this case, the slope is increased and the rms timing error is reduced by the same factor. Neglecting $1/f$ noise, we find for the standard deviation of one output period $T_{\text{out}} = 1/f_{\text{out}}$

$$\sigma_{T_{\text{out}}} = \sqrt{2}\,\sigma_t = \frac{1}{2\pi f_{\text{out}}} \sqrt{\frac{F}{\text{SNR}_{\text{in}}}} \tag{40}$$

where we exploited the fact that two successive output periods are uncorrelated for white noise sources. From the error propagation law we obtain the standard deviation of the output frequency

$$\sigma_{f_{\text{out}}} = \sigma_{T_{\text{out}}} f_{\text{out}}^2 = \frac{f_{\text{out}}}{2\pi} \sqrt{\frac{F}{\text{SNR}_{\text{in}}}} . \tag{41}$$

Averaging n frequencies within one sweep reduces the standard deviation by a factor of $1/\sqrt{n} = 1/\sqrt{T_s f_{\text{out}}}$. Using (41) we obtain the standard deviation of the IF frequency averaged over one sweep given by

$$\sigma_{\overline{f_{\text{out}}}} = \frac{1}{2\pi} \sqrt{\frac{f_{\text{out}} F}{T_s\,\text{SNR}_{\text{in}}}} . \tag{42}$$

The output frequency can be expressed by the sweep bandwidth b_s and the target distance R according to

$$f_{\text{out}} = \frac{2\,b_s\,R}{T_s\,c_0} . \tag{43}$$

Using error propagation again, we obtain from (43) and (42)

$$\sigma_R = \frac{T_s\,c_0}{2\,b_s} \sigma_{\overline{f_{\text{out}}}} = \frac{1}{2\pi} \sqrt{\frac{c_0\,F\,R}{2\,b_s\,\text{SNR}_{\text{in}}}} . \tag{44}$$

From the radar range equation we know that the received signal power is proportional to $1/R^4$, if near-field effects are disregarded. Therefore, we can write the SNR as

$$\text{SNR}_{\text{in}} = snr/R^4 \tag{45}$$

where snr is a normalized SNR at the receiver input. Substituting (45) into (44), we obtain

$$\sigma_R = \frac{1}{2\pi} \sqrt{\frac{c_0\,F}{2\,b_s\,snr}}\, R^{5/2} . \tag{46}$$

Note that the dependence $\sigma_R \propto R^{5/2}$ is stronger than for the distance error from PLL phase noise, where $\sigma_R \propto R^m$ with $1 \le m < 1.5$. As with the PLL phase noise contribution to the distance error, we have disregarded the cross-correlation between different IF periods. An inclusion of these contributions, e. g., by an additional evaluation of the sweep time or the signal phase, respectively, would reduce the noise by a factor $n^{-1/2} = \sqrt{c_0/(2b_s R)} \propto R^{-1/2}$. More accurate expressions for the CRLB for this case are given in [29].

Summarizing this section, for white receiver noise the rms distance error in the far field obeys $\sigma_R \propto R^m$ with $2 \le m < 2.5$.

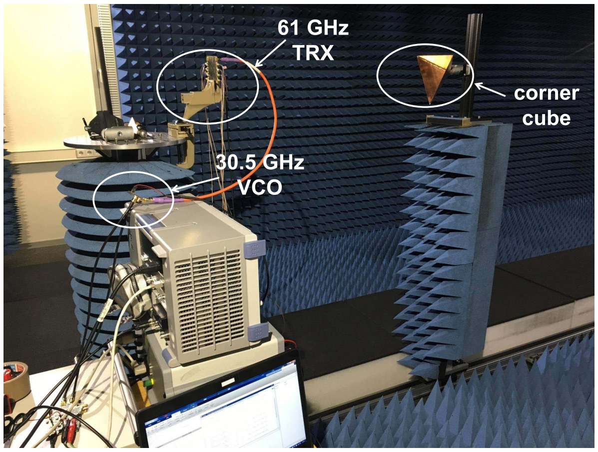

Fig. 6. Measurement setup consisting of a 61 GHz transceiver and a large corner cube used as a target placed on a linear positioner.

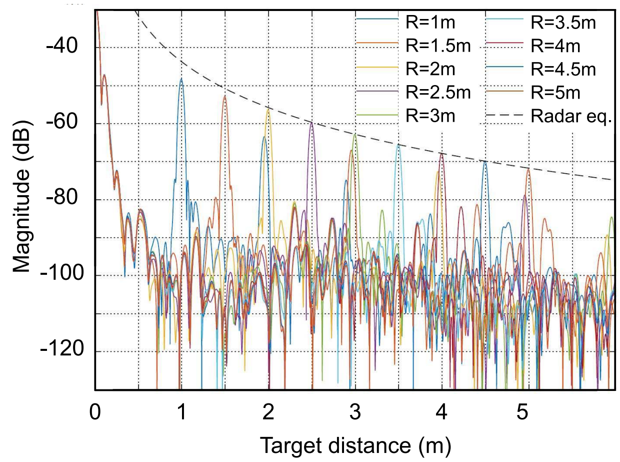

Fig. 7. Measured radar spectra for different target distances R.

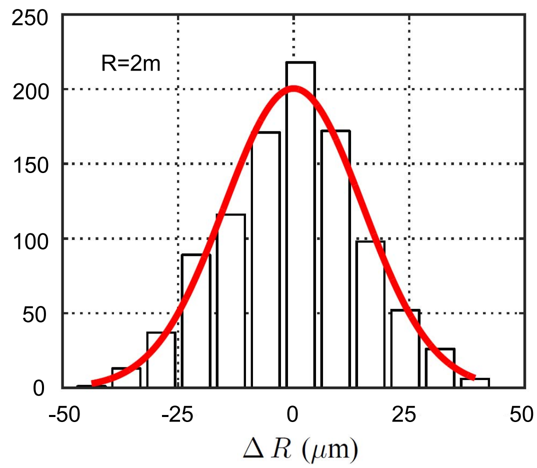

Fig. 8. Histogram for the measured error values at $R = 2$ m for the 61 GHz system. Sweep duration is $T_s = 0.5$ ms and noise bandwidth is $b_n = 2$ MHz.

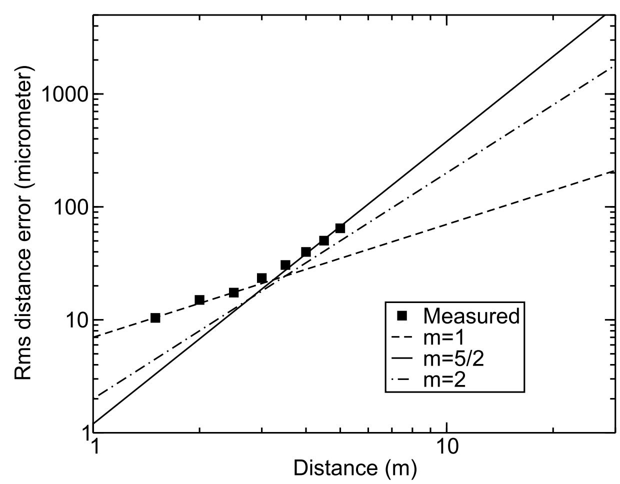

Fig. 9. Measured distance error at 61 GHz (symbols) and fitted curves with $\sigma_R \propto R^m$. Sweep duration is $T_s = 0.5$ ms and noise bandwidth is $b_n = 2$ MHz.

VII. MEASUREMENTS

We performed precision measurements on a scalable sensor platform with 61 and 122 GHz transceivers described in [9]. The sweep bandwidth b_s was as large as 5 GHz for the 61 GHz system and 10 GHz for the 122 GHz system. The distance measurement is based on the evaluation of the FFT spectrum of the IF signal. For a fast chirp of 5 GHz/0.5 ms, the ADC sampling frequency is 10 MHz, which results in 5000 samples. The data samples are zero padded to 2^{25} in FFT. For a slow chirp of 5 GHz/4 ms, the ADC sampling frequency is 1 MHz, which results in 4000 samples. The data samples are zero padded to 2^{25} in FFT. As discussed in Section V-A, the auto-correlation time is much shorter than the IF period and the cross-correlation between adjacent IF periods is insufficiently taken into account if the FFT evaluation is restricted to the amplitude. For this case, (27) may be good approximation for the phase noise induced distance error as discussed at the end of Section V-A and in the appendix.

Fig. 6 shows the setup for the range measurement of a static target in an anechoic chamber. A corner cube was used as a target on a linear positioner. The distance of the target was varied from 1 m to 5 m. Fig. 7 shows the magnitude of the radar signal for the 61 GHz radar system with different target distances. A fit $\propto 1/R^4$ according to the well-known radar range equation is also shown. As evident from Fig. 7, the magnitude accurately follows the radar equation for distances above 2 m. This means that the relationships (45) and (46) are expected to be accurate for $R > 2$ m.

In order to investigate the white noise in the receiver, we used a large noise bandwidth of 2 MHz and a relatively small sweep time of 0.5 ms. The distance measurements were repeated 1000 times. A Gaussian distribution was then fitted to the error distribution. Fig. 8 shows the resulting histogram for the 61 GHz radar system for a target distance of 2 m. The rms distance error as a function of the target distance is shown in Fig. 9. Remember that (46) predicts $\sigma_R \propto R^{5/2}$ in the far field for white noise in the receiver. If the cross-correlation is included, we obtain $\sigma_R \propto R^2$. In our case, the slope m is slightly larger than two for $R > 3$ m. In our measurements a reference frequency as low as 25 MHz was used, which raises the in-band phase noise by about 12 dB compared to the recommended input frequency of 100 MHz. This facilitates precision measurements resulting from phase noise. The choice of this low input frequency was due to the fact that the radar frontend has a relatively low divider output frequency slightly below 2 GHz. The minimum division ratio

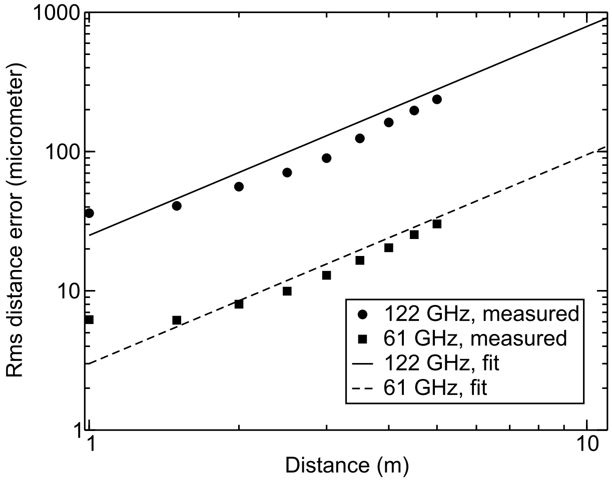

Fig. 10. Measured distance error at 122 GHz and 61 GHz (symbols) and fitted curves with $\sigma_R \propto R^{3/2}$. Sweep duration is $T_s = 4$ ms and noise bandwidth is $b_n = 100$ kHz.

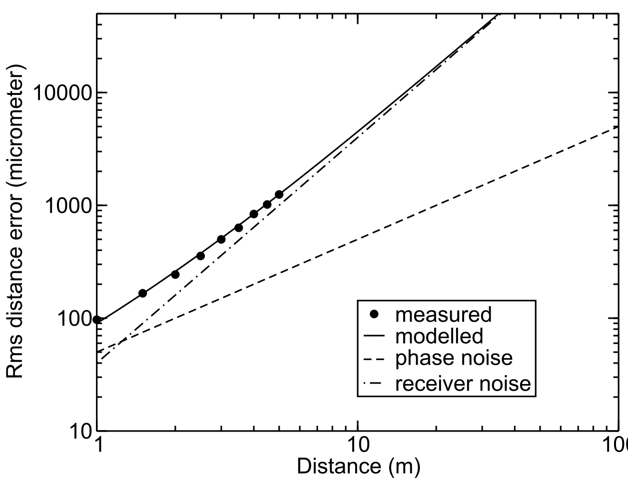

Fig. 11. Measured distance error 122 GHz with small corner cube (symbols) and modelled curve (solid line), PLL phase noise contribution (dashed line) and receiver noise contribution (dot-dashed line). Sweep duration is $T_s = 4$ ms and noise bandwidth is $b_n = 100$ kHz.

of the programmable divider then forced us to use such a low reference frequency. Using a higher reference frequency for the PLL would reduce the distance error mainly for $R < 3$ m, where it is phase-noise limited now.

In order to investigate the effect of in-band phase noise, we used a larger sweep time of 4 ms and a lower noise bandwidth of $b_n = 100$ kHz. Fig. 10 shows the resulting distance error as a function of the target distance R. Since the IF radar signal and the noise bandwidth are below the PLL loop bandwidth we expect a dependence $\sigma_R \propto R^{3/2}$ according to (34). For $R > 2.5$ m this relationship is approximately fulfilled as evident from Fig. 10. This indicates that the distance error is dominated by the in-band PLL phase noise. In order to make the receiver noise visible for this case, we used a smaller corner cube to reduce the received power. This reduces the SNR at the receiver input and enhances the receiver noise contribution to the distance error. Fig. 11 shows the rms distance error as a function of the target distance measured with the smaller corner cube, again for a noise bandwidth of $b_n = 100$ kHz. The solid line represents the modeled dependence according

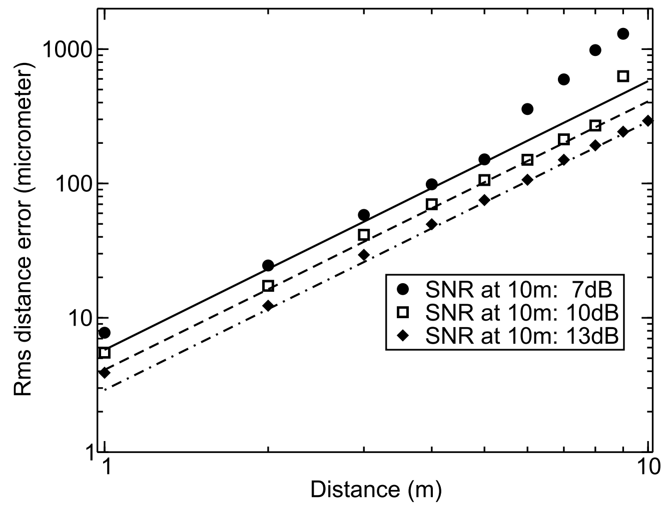

Fig. 12. Simulated RMS distance error for three different SNR values at the receiver output. The lines represent fitted curves with the logarithmic slope of $m = 2$.

to $\sigma_R = a\,R + b\,R^2$, where a and b were fitted to the measured values. The first term represents the PLL phase noise contribution and the second term the white noise in the receiver. For small distances, the phase noise increases roughly in proportion to R, since phase noise dominates the distance error. At a larger target distance the slope m starts to increase as expected due to receiver noise. At large distances, receiver noise dominates the distance error. This contrasts with the results in [10]. We attribute this difference partly to the strong dependence of the receiver input power on the target distance ($\propto 1/R^4$) in our system visible in Fig. 7, which may not be the case for a larger reflector.

Finally, we come back to the case where not only the amplitude information of the FFT but also the phase information is evaluated. According to the appendix, this reduces the rms distance error by the factor $1/\sqrt{n} = \sqrt{c_0/(2b_s R)} \propto R^{-1/2}$. In this case, the distance error due to PLL phase noise increases by 3-6 dB with doubling the target distance and the white noise in the radar receiver results in an increase by 12 dB in the far field with distance doubling. In order to illustrate this point, we simulated the rms distance error in the time domain including white additive noise. Here, we performed 1000 simulations with $N = 4096$ samples for each target distance. We assumed a sweep bandwidth, b_s, of 5 GHz and a sweep time, T_s, of 500 μs. The mean IF period was estimated from the simulated total sweep time divided by the number of output periods as described in the appendix. Frequency error and distance error were then obtained by using the error propagation law according to (23) and (26), respectively. Fig. 12 shows the numerical result for three different SNR values. As evident from Fig. 12 the error is roughly proportional to R^2 as expected as long as the SNR is at least 13 dB. Note that white noise does not exist in reality. While the voltage PSD *before* the LPF may be considered approximately white, the PSD after the LPF cannot, since the white receiver noise at the mixer output is shaped by the LPF. This reduces the noise bandwidth from the Nyquist frequency $(N-1)/(2\,T_s)$ to the receiver noise bandwidth b_n. Disregarding a possible signal attenuation by the LPF, we obtain for the SNR before the LPF

$$\frac{1}{SNR} = \frac{1}{SNR_{\text{out}}} \frac{N-1}{2\, b_n\, T_s} \approx \frac{1}{SNR_{\text{out}}} \frac{N}{2\, b_n\, T_s}\,. \tag{47}$$

By substituting this expression into (5) of [29] we obtain for the CRLB of the distance error

$$\left(\sigma_R^{\text{CRLB}}\right)^2 = \frac{(3/2)\, c_0^2}{(2\pi)^2\, SNR_{\text{out}}\, b_s^2\, b_n\, T_s}\,. \tag{48}$$

Using $b_n = (b_n/f_{\text{out}})f_{\text{out}} = (b_n/f_{\text{out}}) \times (2b_s R)/(T_s c_0)$ and $n = (2b_s R)/c_0$, the estimate (44) can be expressed by the CRLB as

$$\sigma_R = \frac{n}{\sqrt{6}} \left(\frac{b_n}{f_{\text{out}}}\right)^{1/2} \sigma_R^{\text{CRLB}}\,. \tag{49}$$

Obviously, if the noise bandwidth, b_n, is not much larger than the IF frequency, f_{out}, and the number of periods, n, is moderate, then the estimate is relatively close to the CRLB.

VIII. Conclusion

We have presented a noise model for homodyne radar systems including PLL phase noise and white noise in the receiver. Based on a simple PLL model, the phase noise of the radar signal was modelled in the time domain as a non-stationary stochastic process. For the stationary case, the effect of VCO phase noise on the spectrum of the downconverted radar signal was calculated analytically in the frequency domain. The standard deviation in a distance measurement due to PLL phase noise was expressed as a function of basic PLL parameters, target distance and receiver bandwidth. Special cases were discussed, where radar signal and noise bandwidth are either in the VCO dominated region of the phase noise spectrum or in the in-band phase noise floor. From the rms phase error we calculated the standard deviation in an FMCW radar distance measurement. We have also investigated the influence of white noise in the radar receiver on the distance precision. The distance error σ_R due to receiver noise was found to depend stronger on the target distance R than the phase noise induced error. We have measured the rms distance error, σ_R, for a 61 GHz and a 122 GHz radar system. In accordance to theory, we found $\sigma_R \propto R^m$ where $1 \le m < 1.5$ for small distances due to PLL phase noise and $2 \le m < 2.5$ for large distances due to white additive receiver noise.

Appendix
Estimation of the IF Frequency From One FMCW Sweep

We consider n IF periods within one sweep in the steady state. As discussed in the context of (22), we disregard the cross-correlation between different zero crossings t_i and t_j ($i \neq j$). The deviation of the i-th minus-to-plus zero crossing of the signal from the ideal value is denoted as $\tau_i = t_i - t_i^{\text{ideal}}$, see Fig. 2. The timing error of the i-th period is then given by

$$(\Delta T)_i = \tau_{i+1} - \tau_i \tag{50}$$

and the timing error for the whole sweep reads

$$(\Delta T)_{\text{tot}} = \sum_{i=0}^{n-1} (\Delta T)_i = \sum_{i=0}^{n-1} (\tau_{i+1} - \tau_i) = \tau_n - \tau_0\,. \tag{51}$$

Its variance

$$< (\Delta T)^2_{\text{tot}} >= \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} < (\Delta T)_i (\Delta T)_j > \tag{52}$$

contains n diagonal terms ($i = j$) with $< (\Delta T)_i (\Delta T)_i >= 2\sigma_t^2$, as well as 2(n-1) terms with $|i - j| = 1$ and $< (\Delta T)_i (\Delta T)_j >= -\sigma_t^2$. The negative cross-correlation of these adjacent periods is due to the fact that they share one zero crossing. All cross-correlation terms $< (\Delta T)_i (\Delta T)_j >$ with $|i - j| > 1$ are zero according to our assumption of uncorrelated zero crossings. The summation yields

$$< (\Delta T)^2_{\text{tot}} >= n\, 2\sigma_t^2 - 2(n-1)\, \sigma_t^2 = 2\, \sigma_t^2 \tag{53}$$

and the corresponding standard deviation is given by

$$\sigma_{\Delta T} = \sqrt{2}\, \sigma_t\,. \tag{54}$$

Obviously, (54) can directly be obtained from (51), since τ_0 and τ_n are uncorrelated. In other words, the measurement of the first and the last zero crossing results in the same error as the inclusion of the cross-correlation terms, provided that the number of IF periods, n, is known. In this case, we obtain the Cramér-Rao lower bound for the average period of the sweep given by

$$\sigma_T^{\text{CRLB}} = \frac{\sqrt{2}\, \sigma_t}{n} \tag{55}$$

where $SNR \gg 1$ is assumed. By contrast, neglecting the cross-correlation between different periods the averaging procedure over n statistically independent periods results in

$$\sigma_T = \frac{\sqrt{2}\, \sigma_t}{\sqrt{n}}\,. \tag{56}$$

Comparison of (55) and (56) confirms our former result (29) for the distance error.

In summary, the measurement of the timing error of the sweep, $\sigma_{\Delta T}$, and the number of IF periods, n, results in the same distance error as the inclusion of the cross-correlation. Disregarding the cross-correlation raises the error by a factor of $\sqrt{n}$.

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